

Yesterday we saw in 2D

$$\vec{F} \text{ conservative} \Rightarrow \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

Note: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$ and domain of $\vec{F}$
simply connected (no gaps)

$\Rightarrow \vec{F}$ conservative. Quick check!

For a vector field $\vec{F}: U \rightarrow \mathbb{R}^3$ $\vec{F} = (P, Q, R)$

with P, Q, R having continuous partial derivatives, then

$$\text{curl}(\vec{F}) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$
$$= \begin{pmatrix} \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \\ \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \\ \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \end{pmatrix}$$

measure of rotation!

$$\text{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

note:
also a vector
field!

DIY: check that if $f: U \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ is of class C^2
then $\text{curl}(\nabla f) = \vec{0}$

so if $\vec{F}$ is cons $\Rightarrow \text{curl}(\vec{F}) = \vec{0}$.

If $\text{curl}(\vec{F}) = \vec{0}$ and F 's domain is simply connected
then $\vec{F}$ is cons.

For $\vec{F}: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ w/ $F = (P, Q)$ we can write
 $\vec{F} = (P(x, y), Q(x, y), 0)$ so

$$\text{curl}(\vec{F}) = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}$$

I could, thus, rewrite $\oint_C \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$

$$\text{as } \oint_C \vec{F} \cdot d\vec{r} = \iint_D \underbrace{\text{curl}(\vec{F}) \cdot \hat{k}}_{\text{extracts 3rd comp!}} dx dy$$

where $\vec{F}(x, y) = (P, Q)$
we've now going to elevate this to 3D

→ what if C exists in $\mathbb{R}^3$ and I want
to $\int_C \vec{F}(x,y,z) \cdot d\vec{r}$ along that curve. C

Could, of course do the line integral!

OR, Stokes' theorem.

as with greens, we'll use a surface that has
 C as its boundary!

Thm: Let S be a piecewise regular, oriented surface whose oriented boundary C is a piecewise regular, simple closed curve. If $\vec{F} = (P, Q, R)$ is a continuously diff'able 3D v-f whose domain is a region in $\mathbb{R}^3$ that contains S , then

$$\oint_C \vec{F}(x, y, z) \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n}(x, y, z) dS$$

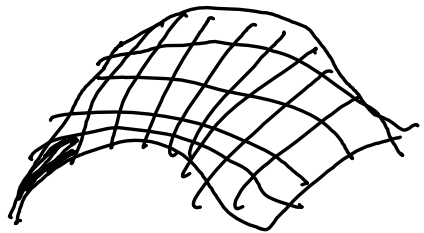
where $\vec{n}(x, y, z)$ is the unit normal vector to S

oriented bd C : counterclockwise.

oriented surface: oriented so that its bd

is counterclockwise - use the "correct" side.

$\iint_S ds$??? surface integrals.



integrate along the surface
not a volume, not a curve,
the surface

For a parametric surface S described by vector f^h
 $\vec{r}(s,t) = (r_1(s,t), r_2(s,t), r_3(s,t)) \quad (s,t) \in D$

check the derivation in the notes:

$$A(S) = \iint_D \|\vec{r}_s(s,t) \times \vec{r}_t(s,t)\| \, ds \, dt$$

(remember, $\|\vec{r}_s \times \vec{r}_t\|$ is area of paral. gm spanned by $\vec{r}_s$ and $\vec{r}_t$)

So, for an arb vector field $\vec{F}(x, y, z)$, I

can find

$$\iint_S \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS$$

flux
integral
of $\vec{F}$ across
 S

$$= \iint_D \vec{F}(\vec{r}(s, t)) \cdot \frac{\vec{r}_s(s, t) \times \vec{r}_t(s, t)}{\|\vec{r}_s(s, t) \times \vec{r}_t(s, t)\|} \|\vec{r}_s(s, t) \times \vec{r}_t(s, t)\| ds dt$$

$$= \iint_D \vec{F}(\vec{r}(s, t)) \cdot \underbrace{(\vec{r}_s(s, t) \times \vec{r}_t(s, t))}_{\text{not unitized!}}$$

which is fine!

of course, we could also want to add up
scalar field on the surface so

$$\iint_S f(x, y, z) dS = \iint_D f(\vec{r}(s, t)) \|\vec{r}_s(s, t) \times \vec{r}_t(s, t)\| ds dt$$

What about other descriptions of S ?

if S is graph of f^h $g: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ w/
cont. 1st order partial derivs then, recall the
normal is given by $(-g_x(x,y), -g_y(x,y), 1)$

$$\text{then } \iint_S \vec{F}(x,y,z) \cdot \vec{n}(x,y,z) dS$$

$$= \iint_D \vec{F}(x,y, \underset{\uparrow}{g(x,y)}) \cdot (-g_x(x,y), -g_y(x,y), 1) dx dy$$

so we're on the surface

and

$$\iint_S f(x,y,z) dS =$$

mag of normal!

$$\iint_D f(x,y, g(x,y)) \sqrt{(g_x(x,y))^2 + (g_y(x,y))^2 + 1} \quad \text{check}$$

note: D is projection of S on xy -plane

Okay so! $\oint_C \vec{F}(x, y, z) \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot \vec{n}(x, y, z) dS$

e.g. $\vec{F}(x, y, z) = (-3y, 2z, -x)$ eval.

$\oint_C \vec{F} \cdot d\vec{r}$ where C is the intersection of plane

$z = y - 1$ w/ cylinder $x^2 + y^2 = 4$

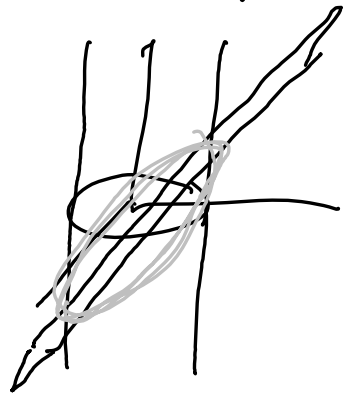
we could describe C by

$$\vec{r}(t) = (2\cos t, 2\sin t, 2\sin t - 1)$$

$$0 \leq t \leq 2\pi$$

$$\int_0^{2\pi} (-6\sin t, 4\sin t - 2, -2\cos t) \cdot (2\cos t, 2\sin t, 2\sin t - 1) dt$$

trig $\cap$



or, let's find a nice easy S
 part of $z=y-1$ enclosed by
 C ; with domain D restricted
 i.e. $2x^2 + y^2 = 4$ — D



$$z = y - 1 \quad \text{normal: } (0, -1, 1)$$

$$g(x, y) = y - 1$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D \text{curl}(\vec{F}) \cdot \vec{n} \, dS = \iint_D \text{curl}(\vec{F}) \cdot (0, -1, 1) \, dx \, dy$$

curl $\vec{F}$:

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -3y & 2z & -x \end{vmatrix} = (0 - 2, -(-1 - 0), 0 - -3) = (-2, 1, 3)$$

$$\iint_D (-2, 1, 3) \cdot (0, -1, 1) \, dx \, dy = \iint_D 2 \, dx \, dy = 2 \iint_D dx \, dy =$$

$$= 2 \cdot (\pi r^2) = 2 \cdot (\pi \cdot 4) = 8\pi \quad \text{or}$$

$$2 \int_0^{2\pi} \int_0^2 r \, dr \, d\theta = 2 \int_0^{2\pi} \frac{r^2}{2} \Big|_0^2 \, d\theta = 2 \int_0^{2\pi} 2 \, d\theta = 4 \cdot 2\pi = 8\pi$$